

Boston Housing Price Prediction using Linear Regression

Submitted By

Name: Sagar Kumar Barick

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Introduction

The objective of this project is to predict housing prices using the Boston Housing Dataset. Linear Regression is applied to understand how different housing features influence house prices and to build a predictive model.

Part A – Theoretical Questions

Q1. What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions or decisions without being explicitly programmed.

Types of Machine Learning

1. Supervised Learning

Uses labeled data for training.

Examples:

- House Price Prediction
- Email Spam Detection

2. Unsupervised Learning

Uses unlabeled data to discover patterns.

Examples:

- Customer Segmentation
- Market Basket Analysis

3. Reinforcement Learning

Learns through rewards and penalties.

Examples:

- Self-driving Cars
- Game Playing AI

Q2. What is Supervised Learning?

Supervised Learning is a machine learning technique where the model learns from labeled training data.

Regression vs Classification

Regression	Classification
Predicts continuous values	Predicts categories
Example: House Price	Example: Spam/Not Spam
Output: Numeric	Output: Class Label

Q3. What is Linear Regression?

Linear Regression is a statistical technique used to model the relationship between independent variables and a dependent variable.

Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where:

- Y = Target Variable
- β_0 = Intercept
- $\beta_1 \dots \beta_n$ = Coefficients
- $X_1 \dots X_n$ = Features

Intercept

The expected value of Y when all features are zero.

Coefficient

Represents the change in Y for a one-unit increase in the feature.

Q4. Evaluation Metrics

SSE (Sum of Squared Errors)

Measures total squared prediction errors.

MSE (Mean Squared Error)

Average of squared prediction errors.

RMSE (Root Mean Squared Error)

Square root of MSE.

R^2 Score

Measures proportion of variance explained by the model.

Adjusted R^2

Modified R^2 considering number of predictors.

Q5. Machine Learning Workflow

1. Data Collection
 2. Data Cleaning
 3. Exploratory Data Analysis
 4. Feature Selection
 5. Train-Test Split
 6. Model Building
 7. Model Evaluation
 8. Model Deployment
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Part B – Practical Implementation

Q6. Load Dataset and Explore Data

Python Code

```
import pandas as pd

df = pd.read_csv("BostonHousing.csv")

print(df.head())
print(df.info())
print(df.describe())
```

Observation

The dataset contains housing-related variables such as crime rate, number of rooms, tax rate, and house price (MEDV).

Q7. Exploratory Data Analysis

Pairplot

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(df[['rm', 'lstat', 'ptratio', 'medv']])
plt.show()
```

Heatmap

```
plt.figure(figsize=(12,8))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```

Correlation Findings

Strong Positive Correlation:

- RM ↔ MEDV

Strong Negative Correlation:

- LSTAT ↔ MEDV
- PTRATIO ↔ MEDV

Q8. Feature and Target Selection

```
X = df.drop('medv', axis=1)
y = df['medv']
```

Target Variable:

- MEDV

Independent Variables:

- All remaining columns

Q9. Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42
)
```

80% Training Data

Q10. Linear Regression Model

Training

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

Prediction

```
y_pred = model.predict(X_test)
```

Evaluation

```
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = mse ** 0.5
r2 = r2_score(y_test, y_pred)

n = len(y_test)
p = X_test.shape[1]

adjusted_r2 = 1 - ((1-r2)*(n-1)/(n-p-1))
```

Output Metrics

Report the values obtained after execution:

- MAE = _
- MSE = _
- RMSE = _
- R^2 = _
- Adjusted R^2 = _

Regression Equation

After training:

```
print(model.intercept_)  
print(model.coef_)
```

General Equation:

$$\text{MEDV} = \beta_0 + \beta_1(\text{CRIM}) + \beta_2(\text{ZN}) + \beta_3(\text{INDUS}) + \beta_4(\text{CHAS}) + \beta_5(\text{NOX}) + \beta_6(\text{RM}) + \beta_7(\text{AGE}) + \beta_8(\text{DIS}) + \beta_9(\text{RAD}) + \beta_{10}(\text{TAX}) + \beta_{11}(\text{PTRATIO}) + \beta_{12}(\text{B}) + \beta_{13}(\text{LSTAT})$$

Interpretation of Coefficients

RM

Positive coefficient indicates that more rooms increase house prices.

LSTAT

Negative coefficient indicates that higher lower-status population percentage decreases house prices.

PTRATIO

Negative coefficient suggests higher student-teacher ratio reduces property value.

CRIM

Higher crime rates generally reduce housing prices.

Conclusion

A Linear Regression model was successfully built using the Boston Housing Dataset. Exploratory Data Analysis revealed significant relationships between housing characteristics and property values. The model achieved satisfactory prediction performance and identified important factors such as number of rooms, crime rate, and lower-status population percentage as major determinants of house prices.